

L10b: The NPV of Composite Contracts and A New Role for Delta

CHEME 5660 Cornell University

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Objectives for Today

Today, we work two examples with the Greeks, the POE, and the POP, then combine calls and puts into composite contracts. What is a composite contract worth, and when does it make money?

Today's concepts

- The NPV and the probability of profit of a composite contract: sums over its legs.
- Common composite contracts at expiration: outlook, breakevens, maximum profit and loss, and the cost of closing early.
- Delta-neutral positions: the net Delta of options and shares, and the shares that make it zero.

What You Need to Know

Core results

- Profit at expiration: the legs' payoffs, weighted by $\theta_i n_i$, minus the net premium grown to expiration. POP: the probability it is positive.
- Read a profit diagram: the breakevens, the maximum profit, and the maximum loss. For the eight credit contracts: the maximum profit is the credit grown to expiration, and the maximum loss comes from the put or call side.
- Composite Delta: $\Delta_{\mathcal{C}} = \sum_{i \in \mathcal{C}} \theta_i n_i \Delta_i$ (per share). Net Delta: $N_S + \Delta_{\text{opts}}$, with $\Delta_{\text{opts}} = m\,\Delta_{\mathcal{C}}$. Holding $N_S = -\Delta_{\text{opts}}$ shares makes it delta-neutral.

For understanding (you won't reproduce these)

- The NPV derivation, and the formula tables for the eight composite contracts (reference).

Lectures and Examples

Lecture:
CHEME-5660-L10b-Lecture-POP-POE-Delta-CompositeContracts-Fall-2026.ipynb

Example 1:
CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Example 2:
CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Example 3:
CHEME-5660-L10b-Example-Delta-VerticalSpread-Fall-2026.ipynb

Review on your own:
CHEME-5660-L10b-Example-Straddle-AMD-Earnings-Fall-2026.ipynb

Review: The Greeks

For small changes in the share price, the volatility, the calendar time (years), and the rate, the premium $\mathcal{P}_\star$ of a call ($\star = c$) or put ($\star = p$) changes by:

$$\Delta\mathcal{P}_\star \approx \Delta_\star\,\Delta S + \tfrac{1}{2}\,\Gamma_\star\,(\Delta S)^2 + \nu_\star\,\Delta\sigma + \Theta_\star\,\Delta t + \rho_\star\,\Delta g_y$$

- Delta and Gamma: the slope and curvature of the premium in S.
- Vega, Theta, and Rho: sensitivities to the volatility σ, the calendar time t, and the risk-free rate g_y, quoted as $\Theta_\star/365$ per day and $\nu_\star/100$, $\rho_\star/100$ per point.
- Hedge ratio: one contract covers 100 shares, so it is equivalent to $100 \cdot \Delta_\star$ shares.

Example 1: CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

Review: POE and POP

The POE is the probability of finishing in the money by more than 0.01 USD/share; the POP is the probability of a positive profit at expiration, beyond the breakeven $S^{\ddagger} = K \pm \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_\star$ ($+$ call, $-$ put):

$$\text{POE} = \begin{cases} \mathbb{P}\left(S(T) > K + 0.01\right) & \text{call} \\ \mathbb{P}\left(S(T) < K - 0.01\right) & \text{put} \end{cases}$$

$$\text{POP} = \begin{cases} \mathbb{P}\left(S(T) > S^{\ddagger}\right) & \text{bought a call or sold a put} \\ \mathbb{P}\left(S(T) < S^{\ddagger}\right) & \text{sold a call or bought a put} \end{cases}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$; we simulate $S(T)$ under $\mathbb{P}$ or $\mathbb{Q}$. With no dividends, a short-dated call's Delta is slightly above its POE under $\mathbb{Q}$.

Example 2: CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

Composite Contracts

A composite contract $\mathcal{C}$ combines long and short puts and calls on one underlying asset with one expiration date T. Its payoff per share at expiration is:

$$\hat{V}_{\mathcal{C}}(S(T)) = \sum_{i \in \mathcal{C}} \theta_i \, n_i \, \underbrace{V_i(K_i, S(T))}_{\text{payoff of leg } i}$$

Here $S(T)$ is the share price at expiration, K_i the strike price of leg i, V_i its payoff, $(S(T) - K_i)^+$ for a call or $(K_i - S(T))^+$ for a put, n_i its number of contracts, and $\theta_i = -1$ if short or 1 if long.

The Net Present Value of a Composite Contract

For one contract, $\mathrm{NPV}_0 = -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot V(K, S(T))$, with $\mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y T}$ at the benchmark growth rate g_y. Summing over the legs:

$$\mathrm{NPV}_0(\mathcal{C}) = \sum_{i \in \mathcal{C}} \theta_i \, n_i \Big(-\mathcal{P}_i + \mathcal{D}_{T,0}^{-1}(g_y) \cdot V_i(K_i, S(T)) \Big) \qquad \text{weighted leg NPVs}$$

$$= \underbrace{-\sum_{i \in \mathcal{C}} \theta_i \, n_i \, \mathcal{P}_i}_{= \mathcal{P}_{\mathcal{C}}} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot \underbrace{\sum_{i \in \mathcal{C}} \theta_i \, n_i \, V_i(K_i, S(T))}_{= \hat{V}_{\mathcal{C}}(S(T))} \qquad \text{split the sum}$$

$$= -\mathcal{P}_{\mathcal{C}} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot \hat{V}_{\mathcal{C}}(S(T))$$

where $\mathcal{P}_i$ is the premium of leg i (USD/share) and $\mathcal{P}_{\mathcal{C}}$ is the net premium of the composite contract.

Profit at Expiration

Factoring out $\mathcal{D}_{T,0}^{-1}(g_y)$ writes the NPV as a profit at expiration discounted to today:

$$\begin{aligned}\mathrm{NPV}_0(\mathcal{C}) &= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \underbrace{\left(\hat{V}_{\mathcal{C}}(S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{\mathcal{C}}\right)}_{= \hat{P}_{\mathcal{C}}(S(T))} \qquad \text{factor out } \mathcal{D}_{T,0}^{-1}(g_y) \\ &= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \hat{P}_{\mathcal{C}}(S(T))\end{aligned}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$ is the accumulation factor, and:

$$\boxed{\hat{P}_{\mathcal{C}}(S(T)) = \hat{V}_{\mathcal{C}}(S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{\mathcal{C}}}$$

is the profit at expiration (USD/share). $\mathcal{P}_{\mathcal{C}} > 0$ opens for a debit; $\mathcal{P}_{\mathcal{C}} < 0$ opens for a credit of $-\mathcal{P}_{\mathcal{C}}$.

Probability of Profit and Probability of Exercise

The NPV and the profit at expiration have the same sign, so with $\mathbb{P}$ our belief, the probability of profit is:

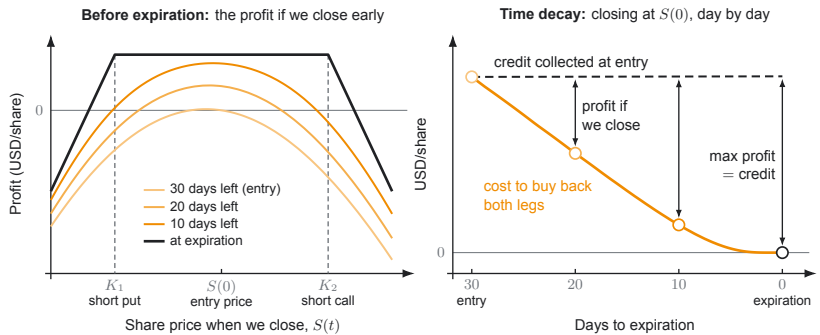
$$\boxed{\text{POP}_{\mathcal{C}} = \mathbb{P}\Big(\text{NPV}_0(\mathcal{C}) > 0\Big) = \mathbb{P}\Big(\hat{P}_{\mathcal{C}}(S(T)) > 0\Big)}$$

Each leg has its own probability of exercise:

$$\text{POE}_i = \begin{cases} \mathbb{P}\left(S(T) > K_i + 0.01\right) & \text{if leg } i \text{ is a call} \\ \mathbb{P}\left(S(T) < K_i - 0.01\right) & \text{if leg } i \text{ is a put} \end{cases}$$

Why Expiration?

Closing a short leg early means buying it back: intrinsic plus extrinsic value $E_V(t)$. A short strangle (put $K_1 = 90$, call $K_2 = 110$, $S(0) = 100$ USD/share, 30 days, CRR tree, $\sigma = 0.40$, $g_y = 4\%$):



Why Expiration? What Do We See?

- Before expiration (left): closing earns the credit grown to that day minus the cost to buy the legs back. Every earlier curve lies below the profit at expiration.
- At the entry share price (right): closing at once costs exactly the credit, all of it extrinsic value.
- Time decay: as the extrinsic value decays, the profit if we close rises, up to the whole credit at expiration.

An early exit falls short of the profit at expiration by roughly the extrinsic value left in the legs.

Common Composite Contracts

Each contract below opens for a credit and has a put side and a call side:

- Sides: empty, a naked short contract, or a credit spread (a short contract protected by a long one farther out).
- Outlook: losing only if the share price falls is neutral to bullish, only if it rises neutral to bearish, either way neutral.
- Maximum loss: empty, none; credit spread, strike width minus the credit; naked put, strike minus the credit; naked call, unlimited.

Common Composite Contracts: Sides and Outlook

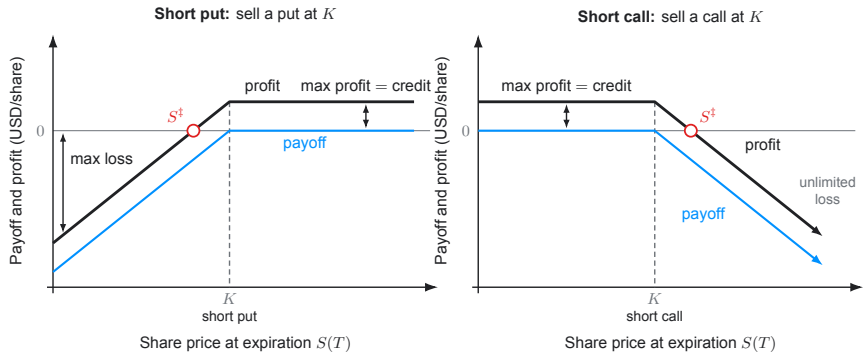
The eight structures, by side:

Composite contract	Put side	Call side	Outlook
Short put	naked short put	none	neutral to bullish
Short call	none	naked short call	neutral to bearish
Put credit spread	put credit spread	none	neutral to bullish
Call credit spread	none	call credit spread	neutral to bearish
Short straddle	naked short put	naked short call at the same strike	neutral
Short strangle	naked short put	naked short call at a higher strike	neutral
Protected strangle	naked short put	call credit spread	neutral, leaning bullish
Iron condor	put credit spread	call credit spread	neutral

Strikes are numbered low to high, $\mathcal{P}_i$ is the premium of the leg at K_i, and $\mathcal{D} = \mathcal{D}_{T,0}(g_y)$. Figures: payoff (blue), profit (black).

Short Puts and Short Calls

Selling one put, a short put, or one call, a short call, is the building block of every structure below:



Each keeps the whole premium if it expires out of the money.

Short Puts and Short Calls: Profit at Expiration

With premiums $\mathcal{P}_p$ (put) and $\mathcal{P}_c$ (call) at strike K, the profits at expiration, and the breakevens $S^\ddagger$, are:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D} \cdot \mathcal{P}_p - (K - S(T))^+ \qquad \text{short put}$$

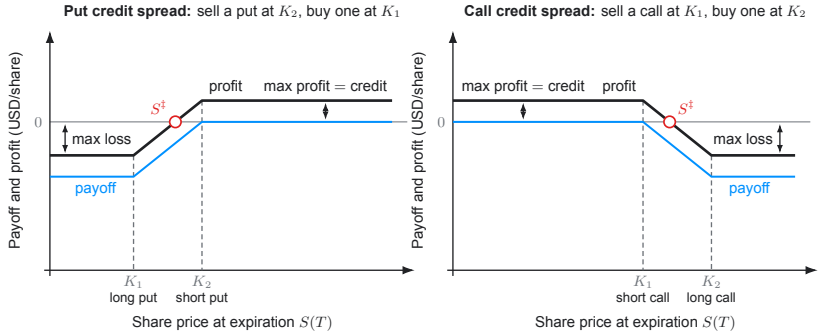
$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D} \cdot \mathcal{P}_c - (S(T) - K)^+ \qquad \text{short call}$$

	Short put	Short call
Maximum profit	$\mathcal{D} \cdot \mathcal{P}_p$ for $S(T) \geq K$	$\mathcal{D} \cdot \mathcal{P}_c$ for $S(T) \leq K$
Breakeven, put side	$S^\ddagger = K - \mathcal{D} \cdot \mathcal{P}_p$	none
Maximum loss, put side	$K - \mathcal{D} \cdot \mathcal{P}_p$ at $S(T) = 0$	none
Breakeven, call side	none	$S^\ddagger = K + \mathcal{D} \cdot \mathcal{P}_c$
Maximum loss, call side	none	unlimited

The short put's loss is bounded ($S(T) \geq 0$); the short call's is not.

Credit Spreads

A credit spread sells the contract closer to the share price and buys one farther away, same type and expiration (a bull put or bear call spread):



The short leg collects the credit; the long leg caps the loss.

Put Credit Spread

A long put at K_1 and a short put at K_2: the credit grown to expiration, minus and plus the leg payoffs:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D} \cdot (\mathcal{P}_2 - \mathcal{P}_1) - (K_2 - S(T))^+ + (K_1 - S(T))^+$$

Maximum profit	$\mathcal{D} \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ for $S(T) \geq K_2$
Breakeven, put side	$S^\ddagger = K_2 - \mathcal{D} \cdot (\mathcal{P}_2 - \mathcal{P}_1)$
Maximum loss, put side	$(K_2 - K_1) - \mathcal{D} \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ for $S(T) \leq K_1$

Call Credit Spread

A short call at K_1 and a long call at K_2 give the profit at expiration:

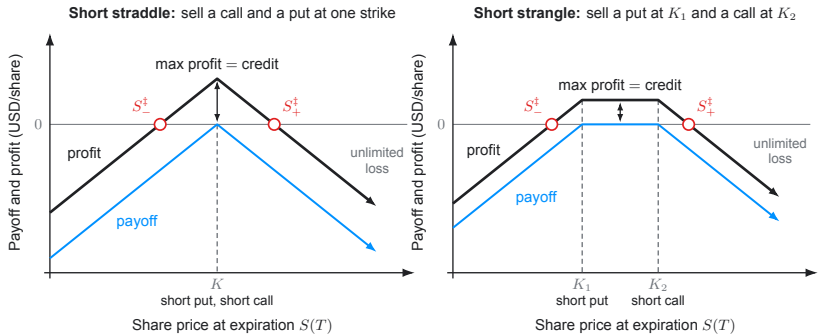
$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D} \cdot (\mathcal{P}_1 - \mathcal{P}_2) - (S(T) - K_1)^+ + (S(T) - K_2)^+$$

Maximum profit	$\mathcal{D} \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \leq K_1$
Breakeven, call side	$S^\ddagger = K_1 + \mathcal{D} \cdot (\mathcal{P}_1 - \mathcal{P}_2)$
Maximum loss, call side	$(K_2 - K_1) - \mathcal{D} \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \geq K_2$

Beyond the long strike, the long leg gains what the short leg loses, so the loss stops at the strike width minus the credit.

Short Straddles and Strangles

A straddle: a put and a call at one strike K. A strangle: a put at K_1 below a call at K_2. Same expiration:



Sold, each profits only between its breakevens: POP $= \mathbb{P}(S_-^\ddagger < S(T) < S_+^\ddagger)$.

Short Straddle

Selling a put and a call at the strike K gives the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D} \cdot (\mathcal{P}_p + \mathcal{P}_c) - (K - S(T))^+ - (S(T) - K)^+$$

Maximum profit	$\mathcal{D} \cdot (\mathcal{P}_p + \mathcal{P}_c)$ at $S(T) = K$
Breakeven, put side	$S_-^\ddagger = K - \mathcal{D} \cdot (\mathcal{P}_p + \mathcal{P}_c)$
Maximum loss, put side	$K - \mathcal{D} \cdot (\mathcal{P}_p + \mathcal{P}_c)$ at $S(T) = 0$
Breakeven, call side	$S_+^\ddagger = K + \mathcal{D} \cdot (\mathcal{P}_p + \mathcal{P}_c)$
Maximum loss, call side	unlimited

The straddle collects more credit, but the strangle keeps its credit over a wider range of share prices.

Short Strangle

Selling a put at K_1 and a call at K_2 gives the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D} \cdot (\mathcal{P}_1 + \mathcal{P}_2) - (K_1 - S(T))^+ - (S(T) - K_2)^+$$

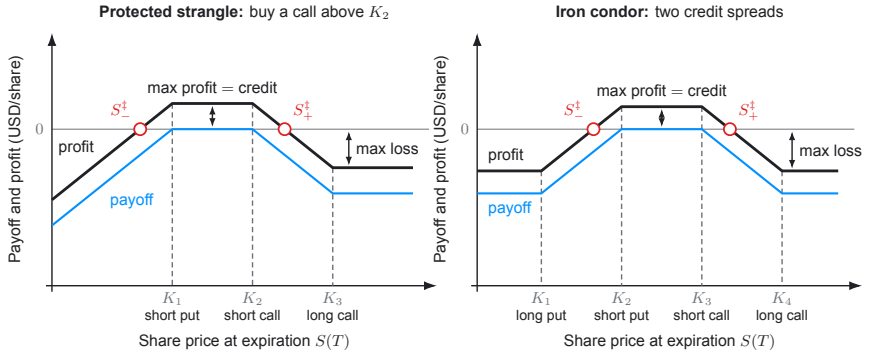
Maximum profit	$\mathcal{D} \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ for $K_1 \leq S(T) \leq K_2$
Breakeven, put side	$S_-^\ddagger = K_1 - \mathcal{D} \cdot (\mathcal{P}_1 + \mathcal{P}_2)$
Maximum loss, put side	$K_1 - \mathcal{D} \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ at $S(T) = 0$
Breakeven, call side	$S_+^\ddagger = K_2 + \mathcal{D} \cdot (\mathcal{P}_1 + \mathcal{P}_2)$
Maximum loss, call side	unlimited

Buying both legs ($\theta_i = 1$) flips either diagram: the loss is capped at the premium paid, and a large move either way pays.

Review on your own: CHEME-5660-L10b-Example-Straddle-AMD-Earnings-Fall-2026.ipynb

Protected Strangles and Iron Condors

A protected strangle (jade lizard) adds a long call at K_3 to a short strangle; an iron condor also adds a long put:



Each long leg caps the loss on its side but lowers the credit.

Protected Strangle

A short put at K_1, a short call at K_2, and a long call at K_3 give the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}\cdot(\mathcal{P}_1+\mathcal{P}_2-\mathcal{P}_3) - (K_1-S(T))^+ - (S(T)-K_2)^+ + (S(T)-K_3)^+$$

protected strangle

Maximum profit	$\mathcal{D}\cdot(\mathcal{P}_1+\mathcal{P}_2-\mathcal{P}_3)$ for $K_1 \le S(T) \le K_2$
Breakeven, put side	$S_-^\ddagger = K_1 - \mathcal{D}\cdot(\mathcal{P}_1+\mathcal{P}_2-\mathcal{P}_3)$
Maximum loss, put side	$K_1 - \mathcal{D}\cdot(\mathcal{P}_1+\mathcal{P}_2-\mathcal{P}_3)$ at $S(T)=0$
Breakeven, call side	$S_+^\ddagger = K_2 + \mathcal{D}\cdot(\mathcal{P}_1+\mathcal{P}_2-\mathcal{P}_3)$[a]
Maximum loss, call side	$(K_3-K_2) - \mathcal{D}\cdot(\mathcal{P}_1+\mathcal{P}_2-\mathcal{P}_3)$ for $S(T)\ge K_3$[b]

[a] If $\mathcal{D}\cdot(\mathcal{P}_1+\mathcal{P}_2-\mathcal{P}_3) \le K_3-K_2$; otherwise none. [b] If $\mathcal{D}\cdot(\mathcal{P}_1+\mathcal{P}_2-\mathcal{P}_3) < K_3-K_2$; otherwise none.

A credit, grown to expiration, of at least K_3-K_2 removes the call-side loss.

Iron Condor

A long put at K_1, a short put at K_2, a short call at K_3, and a long call at K_4 give the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D} \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4) + (K_1 - S(T))^+ - (K_2 - S(T))^+ - (S(T) - K_3)^+ + (S(T) - K_4)^+$$

Maximum profit	$\mathcal{D} \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $K_2 \le S(T) \le K_3$
Breakeven, put side	$S_-^\ddagger = K_2 - \mathcal{D} \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$
Maximum loss, put side	$(K_2 - K_1) - \mathcal{D} \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $S(T) \le K_1$
Breakeven, call side	$S_+^\ddagger = K_3 + \mathcal{D} \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$
Maximum loss, call side	$(K_4 - K_3) - \mathcal{D} \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $S(T) \ge K_4$

The share price finishes beyond at most one side, so the maximum loss is the wider strike width minus the credit.

Delta as a Hedging Tool for Composite Contracts

Delta is also the number of shares that hedges one contract. A composite position's Delta is the sum of its legs' Deltas, so the same idea hedges the whole position.

What is hedging? Offsetting potential losses with an opposite position in a related counter asset, often the underlying shares.

A position with zero Delta is insensitive to small changes in the share price.

What Is the Delta of a Composite Contract?

We hold $\theta_i n_i$ American contracts of leg i, each for m shares, and N_S shares. With $\mathcal{P}_i(S,t)$ the premium of leg i at share price S and time t, the portfolio is worth $\Pi(S,t) = N_S S + m \sum_{i\in\mathcal{C}} \theta_i n_i \mathcal{P}_i(S,t)$, and the net Delta is its sensitivity to the share price:

$$\Delta_{\text{net}}(S,t) = \frac{\partial \Pi}{\partial S} = \underbrace{\frac{\partial (N_S S)}{\partial S}}_{=N_S} + m \sum_{i\in\mathcal{C}} \theta_i n_i \underbrace{\frac{\partial \mathcal{P}_i}{\partial S}}_{=\Delta_i(S,t)}$$

$$= N_S + \underbrace{m \sum_{i\in\mathcal{C}} \theta_i n_i \Delta_i(S,t)}_{=\Delta_{\text{opts}}(S,t)}$$

The options contribute $\Delta_{\text{opts}} = m\,\Delta_{\mathcal{C}}$, where $\Delta_{\mathcal{C}}(S,t) = \sum_{i\in\mathcal{C}} \theta_i n_i \Delta_i(S,t)$ is the composite Delta, per share.

Delta-Neutral Position

A position is delta-neutral if $\Delta_{\text{net}}(S,t) = 0$. We get there by choosing N_S to offset the Delta of the options:

$$\Delta_{\text{net}}(S,t) = N_S + \Delta_{\text{opts}}(S,t) = 0 \quad \Rightarrow \quad N_S = -\Delta_{\text{opts}}(S,t)$$

Quick gut-check example: short 10 calls at $\Delta = 0.6$ with $m = 100$ shares/contract, so $\Delta_{\text{opts}} = 10 \times (-1) \times 0.6 \times 100 = -600$. Holding $N_S = +600$ shares gives $\Delta_{\text{net}} = 600 - 600 = 0$.

If leg i is exercised or assigned at time $\tau \leq T$, set $\Delta_i = 0$ for $t \geq \tau$ and update the shares:

- Call leg i: $N_S \leftarrow N_S + \theta_i \cdot m \cdot n_i$.
- Put leg i: $N_S \leftarrow N_S - \theta_i \cdot m \cdot n_i$.

Delta of the Common Composite Contracts

Short legs at Deltas ∓ 0.25, long legs at ∓ 0.10 beyond them, and a straddle at ∓ 0.5. At entry, one copy of each leg ($m = 100$):

Composite contract	Composite Delta, Δ_c	Shares to hedge, N_S
Short put	$-(-0.25) = 0.25$	-25
Short call	-0.25	$+25$
Put credit spread	$-0.10 - (-0.25) = 0.15$	-15
Call credit spread	$-0.25 + 0.10 = -0.15$	$+15$
Short straddle	$-(-0.5) - 0.5 \approx 0$	≈ 0
Short strangle	$-(-0.25) - 0.25 = 0$	0
Protected strangle	$-(-0.25) - 0.25 + 0.10 = 0.10$	-10
Iron condor	$-0.10 - (-0.25) - 0.25 + 0.10 = 0$	0

At these strikes, the sign of Δ_c matches the outlook.

Reading the Delta Table

$\Delta_{\mathcal{C}}$ is the position's gain per share when the share price rises 1 USD/share:

- The rule of thumb: a Delta of ± 0.25 is roughly a one-in-four chance of finishing in the money under $\mathbb{Q}$.
- The protected strangle: its long call leaves $+0.10$, so it leans bullish.
- Zero is not zero risk: Gamma pulls the Delta away from zero as the share price moves.

Staying delta-neutral means updating N_S as the Deltas change.

Example 3: CHEME-5660-L10b-Example-Delta-VerticalSpread-Fall-2026.ipynb

Summary

Key takeaways

- The NPV of a composite contract is the sum of its legs' NPVs: it is the profit at expiration discounted to today, so the POP is the probability that the NPV is positive.
- Common composite contracts have a put side and a call side: each side is empty, naked, or a credit spread, and sets the outlook and the maximum loss. Closing early pays the extrinsic value left in the legs.
- Delta builds hedged positions: the net Delta is the shares plus the weighted leg Deltas, and holding $N_S = -\Delta_{\text{opts}}$ shares makes it zero.